

ONEDRAFT

CAD-AI — Aria Powered

Billing, Subscription & Entitlement Architecture

Version 1.0

Document: OneDraft Billing, Subscription & Entitlement Architecture Specification

System: OneDraft CAD-AI

Intelligence Layer: Aria

Document Class: Architectural & Implementation Stack Specification

Status: Architectural Specification

Predecessor: Project Lifecycle & State Machine Architecture Specification

Successor: Multi-Tenant Organization & Project Isolation Architecture Specification

1. PURPOSE

The Billing, Subscription & Entitlement Architecture establishes the commercial-control system governing OneDraft plans, subscriptions, usage, quotas, metering, entitlements and organization-level commercial boundaries.

The purpose of this subsystem is to provide a deterministic relationship between commercial configuration and the capabilities available to an organization and its projects.

Billing shall remain separate from core project data, computational correctness, regulatory validation and architectural authority.

A commercial condition shall never alter the authoritative Project Model or invalidate previously generated project records.

The billing architecture shall therefore provide a controlled commercial layer over the OneDraft application while preserving project integrity, data ownership, security and lifecycle continuity.

2. ARCHITECTURAL ROLE

The Billing, Subscription & Entitlement subsystem forms the commercial authorization layer of OneDraft.

It determines which platform capabilities, usage quantities and service resources an organization is entitled to consume according to its active commercial configuration.

Primary responsibility: Manage subscriptions, plans, usage, quotas, metering and feature entitlements.

Primary inputs: Organization identity, subscription configuration, plan definitions, usage events, billing events, administrative configuration and entitlement requests.

Primary outputs: Entitlement decisions, quota states, usage records, billing records, subscription states and commercial events.

Authority: Commercial configuration and subscription state shall be authoritative for entitlement decisions.

Separation: Billing state shall not become authoritative for Project Model, compliance, validation or document correctness.

3. COMMERCIAL ARCHITECTURE

OneDraft shall separate commercial configuration into distinct concepts.

Plan: Defines a commercial product offering.

Subscription: Represents an organization's active commercial relationship with a plan.

Entitlement: Defines a capability or resource available under a subscription.

Quota: Defines a permitted quantity or usage limit.

Meter: Measures consumption.

Usage Record: Records measured consumption.

Billing Period: Defines the period over which usage and subscription activity are evaluated.

Invoice: Represents a billing record generated through the applicable billing system.

Credit: Represents an authorized commercial adjustment.

4. PLAN ARCHITECTURE

A Plan shall define a controlled collection of OneDraft capabilities and commercial limits.

A plan may specify:

Features

Project Limits

Storage Limits

Modeling Capacity

Drawing Generation Capacity

AI Usage

Export Capabilities

API Capacity

User Limits

Organization Limits

Support Level

Billing Interval

Plans shall be versioned so that historical subscriptions remain associated with the plan configuration applicable at the relevant time.

5. PLAN VERSIONING

Plan definitions shall be immutable once used by an active subscription except through controlled versioning.

A new plan configuration shall create a new plan version rather than silently modifying historical commercial conditions.

Historical billing and entitlement decisions shall remain reproducible using the plan version that was active when the decision occurred.

6. SUBSCRIPTION

A Subscription shall represent an organization's commercial enrollment in a OneDraft plan.

A subscription shall contain:

Subscription Identifier

Organization Identifier

Plan Identifier

Plan Version

Status

Start Date

Renewal Date

Billing Interval

Current Period

Cancellation State

Entitlement State

External Billing Reference, where applicable.

7. SUBSCRIPTION STATES

The canonical subscription states shall include:

PENDING

TRIAL

ACTIVE

PAST_DUE

SUSPENDED

CANCELLED

EXPIRED

Subscription state shall be distinct from project lifecycle state.

A project's lifecycle state shall not automatically change merely because a subscription changes state.

8. TRIAL STATE

TRIAL shall permit controlled access to specified OneDraft capabilities for a defined period or usage allowance.

Trial configuration shall specify:

Duration

Included Features

Usage Limits

Project Limits

Storage Limits

Conversion Conditions

Trial usage shall be metered according to the same authoritative usage architecture used for paid subscriptions where practical.

9. ACTIVE STATE

ACTIVE indicates that the subscription is currently entitled to the capabilities defined by its plan and applicable commercial configuration.

An active subscription shall permit entitlement evaluation according to its plan, organization and applicable usage conditions.

10. PAST_DUE STATE

PAST_DUE indicates that a billing condition requires attention while the subscription remains subject to the applicable grace and entitlement rules.

The system shall distinguish between billing delinquency and immediate loss of access.

Grace periods and restricted-access policies shall be configurable.

11. SUSPENDED STATE

SUSPENDED indicates that subscription capabilities have been restricted according to defined commercial policy.

Suspension shall not delete project information.

Existing project artifacts and historical records shall remain subject to their applicable retention and access policies.

12. CANCELLED STATE

CANCELLED indicates that the subscription has been terminated.

Cancellation shall preserve commercial history and shall not inherently delete organization or project data.

The organization may retain access to data according to the applicable plan, contractual configuration and administrative policies.

13. EXPIRED STATE

EXPIRED indicates that a subscription has reached the end of its permitted commercial period without renewal or continuation.

Expiration shall produce a deterministic entitlement state.

The platform shall distinguish expired subscription status from deleted organization or project data.

14. ENTITLEMENT MODEL

An Entitlement defines whether a specific organization or user may consume a particular OneDraft capability.

Representative entitlements include:

Project Creation

Project Modeling

Geometry Computation

Compliance Analysis

Validation

Drawing Generation

CAD Export

BIM Export

AI Interaction

API Access

Artifact Storage

Advanced Review

Administrative Controls

Entitlements shall be evaluated using authoritative subscription and organization information.

15. ENTITLEMENT TYPES

Entitlements may be:

Boolean: Capability is enabled or disabled.

Quantity-Based: Capability is limited to a defined quantity.

Rate-Based: Capability is limited within a defined time interval.

Resource-Based: Capability depends upon available computational or storage resources.

Role-Based: Capability depends upon organizational role.

Project-Based: Capability applies to specified project types or quantities.

16. ENTITLEMENT EVALUATION

An entitlement request shall be evaluated against:

Organization

Subscription

Plan

Plan Version

Feature

Usage

Quota

Project

User

Authorization

Administrative Configuration

The resulting decision shall identify whether the operation is permitted.

17. ENTITLEMENT DECISION

An entitlement decision shall contain sufficient information to explain the result.

Decision: Allowed or denied.

Entitlement: Capability evaluated.

Subject: Organization, user or project.

Reason: Deterministic reason for the decision.

Quota State: Applicable remaining or consumed quantity.

Plan: Relevant commercial configuration.

Timestamp: Time of evaluation.

Entitlement decisions may be logged for operational and audit purposes.

18. QUOTA ARCHITECTURE

Quotas shall establish controlled limits for resource consumption.

Representative quotas include:

Projects Per Organization

Users Per Organization

Storage

AI Requests

AI Compute

Drawing Generation

Geometry Computation

Exports

API Requests

Background Jobs

Quotas shall be configurable by plan and organization.

19. QUOTA PERIODS

Quota measurements may operate over:

Billing Period

Calendar Period

Daily Period

Monthly Period

Project Lifetime

Organization Lifetime

The applicable period shall be explicitly defined by the quota configuration.

20. USAGE METERING

OneDraft shall maintain an authoritative usage-metering architecture.

Usage shall be generated from measurable system events.

Potential usage dimensions include:

AI Tokens

AI Requests

Geometry Compute Time

Drawing Generation Jobs

Export Operations

Storage Consumption

API Requests

Worker Runtime

Project Count

Usage records shall identify the organization and applicable project where appropriate.

21. USAGE EVENTS

Usage shall be captured through structured events.

A usage event shall include:

Usage Event Identifier

Organization Identifier

Project Identifier, where applicable.

Meter Type

Quantity

Unit

Source Event

Timestamp

Actor

Correlation Identifier

Usage events shall be immutable after authoritative recording.

22. USAGE AGGREGATION

Usage events shall be aggregated into authoritative usage totals.

Aggregation shall support:

Current Period Usage

Historical Usage

Project Usage

Organization Usage

Meter Usage

Quota Consumption

Aggregation shall be deterministic and reproducible.

23. IDEMPOTENCY

Usage recording shall support idempotency.

The same underlying operational event shall not create duplicate billable usage merely because a worker, event consumer or API request was retried.

Usage events shall therefore possess unique identifiers or equivalent idempotency keys.

24. BILLING PERIOD

Each subscription shall have a defined billing period.

The billing period shall determine:

Usage Window

Quota Reset

Recurring Charges

Billing Events

Subscription Renewal

The system shall preserve historical billing-period records.

25. INVOICING

Where OneDraft provides or integrates with invoicing functionality, invoices shall represent commercial records associated with the organization and subscription.

An invoice may include:

Subscription Charges

Usage Charges

Credits

Adjustments

Taxes

Billing Period

Invoice Status

External payment-provider identifiers shall be stored as references rather than becoming authoritative project records.

26. PAYMENT PROVIDER INTEGRATION

OneDraft may integrate with external payment and billing providers.

External providers shall handle payment processing where applicable.

OneDraft shall maintain the minimum required commercial state needed to evaluate subscriptions and entitlements.

Payment credentials shall not be stored within the OneDraft application unless explicitly required and appropriately secured.

27. BILLING WEBHOOKS

External billing events shall be processed through controlled webhook or integration endpoints.

Incoming billing events shall be:

Authenticated

Validated

Idempotently Processed

Recorded

Mapped to Organization

Mapped to Subscription

Converted into Internal Commercial Events

External billing events shall not directly mutate arbitrary project records.

28. COMMERCIAL EVENT BUS

Commercial events shall integrate with the OneDraft event architecture.

Representative events include:

SubscriptionCreated

SubscriptionActivated

SubscriptionRenewed

SubscriptionPastDue

SubscriptionSuspended

SubscriptionCancelled

SubscriptionExpired

UsageRecorded

QuotaExceeded

EntitlementChanged

Commercial events shall be versioned and traceable.

29. FEATURE ENTITLEMENTS

Platform features shall be represented through controlled feature identifiers.

Example identifiers may include:

feature.project.create

feature.modeling

feature.compliance

feature.validation

feature.drawing.generate

feature.cad.export

feature.bim.export

feature.aria

feature.api

feature.advanced.review

Feature identifiers shall remain stable while their commercial availability is controlled through versioned configuration.

30. ARIA ENTITLEMENTS

Aria access shall be governed by explicit commercial entitlements where the selected OneDraft plan requires it.

Entitlement controls may govern:

Aria Availability

Request Volume

Model Access

Advanced Reasoning Features

Context Capacity

Generation Capacity

Automation Features

API-Based Aria Access

Commercial limits shall not alter the correctness requirements imposed by the AI architecture.

31. AI USAGE METERING

AI usage shall be metered independently from AI decision quality.

Usage metrics may include:

Requests

Input Tokens

Output Tokens

Model Runtime

Compute Units

Tool Operations

Generated Artifacts

The billing architecture shall consume authoritative usage events from the AI runtime rather than attempting to reconstruct usage from frontend activity.

32. COMPUTATIONAL USAGE

Geometry and computational operations may be metered where commercial plans require capacity controls.

Potential metrics include:

CPU Time

Worker Runtime

Geometry Operations

Model Regeneration

Rendering Operations

Large-Model Processing

Computational usage shall be measured by the execution architecture.

33. STORAGE ENTITLEMENTS

Artifact and object storage consumption may be subject to organizational quotas.

Storage calculations shall distinguish between:

Active Artifacts

Historical Artifacts

Imported Artifacts

Generated Artifacts

Temporary Processing Artifacts

Retention and deletion rules shall remain governed by the Data Lifecycle architecture.

34. API ENTITLEMENTS

API usage may be governed through plan-specific limits.

Controls may include:

Requests Per Minute

Requests Per Hour

Requests Per Billing Period

Concurrent Requests

API Feature Access

API entitlement evaluation shall integrate with the API Gateway and Rate Limiting architecture.

35. ORGANIZATION ADMINISTRATION

Organization administrators shall have access to appropriate commercial controls.

Administrative functions may include:

Plan Inspection

Subscription Inspection

Usage Monitoring

Entitlement Inspection

User Allocation

Project Allocation

Billing Configuration

Payment Configuration

Administrative access shall be controlled through the Identity, Access and Authorization architecture.

36. USAGE DASHBOARD

The frontend shall provide an organization-level usage interface.

The dashboard may display:

Current Plan

Subscription State

Billing Period

Usage

Remaining Quotas

Enabled Features

Usage Trends

Subscription Events

The displayed information shall originate from authoritative commercial services.

37. QUOTA ENFORCEMENT

Quota enforcement shall occur before the protected operation is accepted.

Where a quota is exceeded, the application shall return a deterministic entitlement or quota result.

The frontend shall communicate the condition without exposing internal implementation details.

Quota enforcement shall not rely solely upon frontend counters.

38. SOFT AND HARD LIMITS

Commercial limits may be defined as:

Soft Limit: Usage threshold that generates a warning while permitting continued operation.

Hard Limit: Usage threshold that prevents additional consumption.

The applicable limit behavior shall be defined by plan configuration.

39. OVERAGE

Plans may support controlled overage policies.

Where overage is enabled, the system shall record consumption beyond the included quota and associate the excess usage with the applicable billing configuration.

Where overage is disabled, the applicable operation shall be denied once the hard limit is reached.

40. CREDITS AND ADJUSTMENTS

Authorized commercial credits and adjustments shall be represented as explicit records.

Each adjustment shall identify:

Adjustment Identifier

Organization

Subscription

Amount or Quantity

Reason

Authorized Actor

Timestamp

Source

Credits shall not modify historical usage events.

41. COMMERCIAL AUDIT

Commercial operations shall be auditable.

The system shall retain records of:

Subscription Changes

Plan Changes

Entitlement Changes

Usage

Quota Changes

Credits

Billing Events

Administrative Actions

Historical commercial state shall remain reconstructable.

42. PLAN CHANGES

An organization may change plans through an authorized commercial operation.

Plan changes shall create explicit subscription records and effective dates.

The system shall define whether the change is:

Immediate

End-of-Period

Scheduled

The entitlement service shall evaluate the appropriate plan according to the effective timestamp.

43. DOWNGRADE PROTECTION

A plan downgrade shall not silently destroy project information.

Where a downgrade causes the organization to exceed newly applicable quotas, the system shall establish a controlled restricted state.

Existing project data shall remain subject to applicable access and retention policies.

The system shall clearly identify conditions that must be resolved.

44. UPGRADE

An upgrade shall create the applicable entitlement state according to the effective subscription configuration.

New capabilities shall become available only after the authoritative subscription state confirms the upgrade.

Frontend presentation shall follow the authoritative entitlement state.

45. CANCELLATION PROTECTION

Subscription cancellation shall preserve historical billing and project records.

The system shall clearly distinguish:

Subscription Cancellation

from

Organization Deletion

and

Project Deletion

These are separate operations governed by separate architectures.

46. DATA MODEL

The billing architecture shall support structured commercial entities.

Representative entities include:

Plan

PlanVersion

Subscription

SubscriptionPeriod

Entitlement

Quota

Meter

UsageEvent

UsageAggregate

Invoice

BillingEvent

Credit

CommercialAdjustment

EntitlementDecision

BillingConfiguration

These structures shall integrate with Organization, User, Project and API usage structures without becoming replacements for those domain entities.

47. API ARCHITECTURE

The billing API shall expose controlled commercial resources.

Representative endpoints include:

GET /organizations/{id}/billing

GET /organizations/{id}/subscription

GET /organizations/{id}/entitlements

GET /organizations/{id}/usage

GET /organizations/{id}/quotas

GET /organizations/{id}/billing/history

POST /organizations/{id}/subscription

POST /organizations/{id}/subscription/change

POST /organizations/{id}/subscription/cancel

POST /billing/webhooks

The final API contract shall remain governed by the established OneDraft OpenAPI architecture.

48. SECURITY, TESTING & OPERATIONAL INTEGRITY

Billing and entitlement services shall be protected by the OneDraft identity and authorization architecture.

Testing shall include:

Plan Evaluation

Subscription Transitions

Entitlement Decisions

Quota Enforcement

Usage Metering

Idempotency

Billing Webhooks

Plan Changes

Cancellation

Upgrade

Downgrade

Concurrent Usage

Failure Recovery

Security

Commercial decisions shall be deterministic and observable.

The billing system shall not become a privileged pathway for bypassing project authorization, lifecycle controls or security boundaries.

49. ARCHITECTURAL INVARIANTS

The following invariants shall govern the Billing, Subscription & Entitlement Architecture.

Commercial Separation: Billing state shall remain separate from authoritative project state.

Deterministic Entitlements: Identical commercial state and entitlement inputs shall produce the same entitlement result.

Backend Enforcement: Commercial limits shall not depend solely upon frontend controls.

Historical Integrity: Historical subscription and usage records shall remain preserved.

Idempotent Metering: A single operational event shall not be counted more than once.

No Project Destruction: Commercial state changes shall not silently destroy project information.

Explicit Plan Changes: Upgrades and downgrades shall produce explicit commercial records.

Controlled Access: Entitlement decisions shall remain subject to identity and authorization.

AI Separation: AI usage metering shall not determine AI correctness or validation.

Tenant Association: Commercial records shall remain associated with the correct organization.

Auditability: Commercial state changes shall remain traceable.

Provider Independence: External payment providers shall not become authoritative sources for OneDraft project state.

50. COMPLETION STATUS

The Billing, Subscription & Entitlement Architecture establishes the complete commercial-control layer for OneDraft.

The subsystem defines plans, plan versions, subscriptions, subscription states, entitlements, quotas, usage metering, AI usage, computational usage, storage limits, API limits, billing periods, invoicing, external payment integration, commercial events, upgrades, downgrades, cancellation, credits, overage and commercial auditing.

The architecture establishes a controlled relationship between the OneDraft commercial model and the capabilities available to organizations while preserving the independence and integrity of project data, Project Model state, lifecycle state, validation, documentation and security.

The next architectural layer establishes the formal Multi-Tenant Organization & Project Isolation architecture, defining how organizations, users, projects, resources, quotas and data remain isolated within the OneDraft platform.

Status: ARCHITECTURALLY DEFINED

Next Document: Multi-Tenant Organization & Project Isolation Architecture Specification

Overall Architecture: OneDraft End-to-End System Integration